

Project Design Phase Solution Architecture

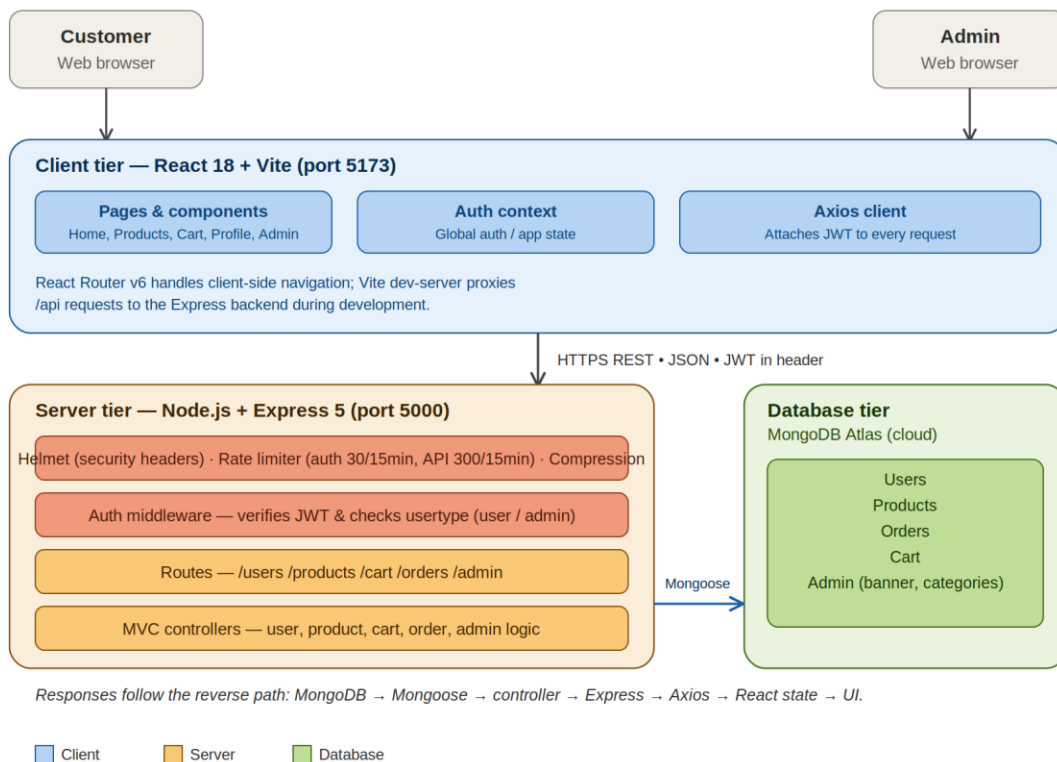
Date	18 June 2026
Name	Dhanish Ladwani
Project Name	ShopEZ – E-Commerce Application
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

ShopEZ – Solution Architecture Diagram:



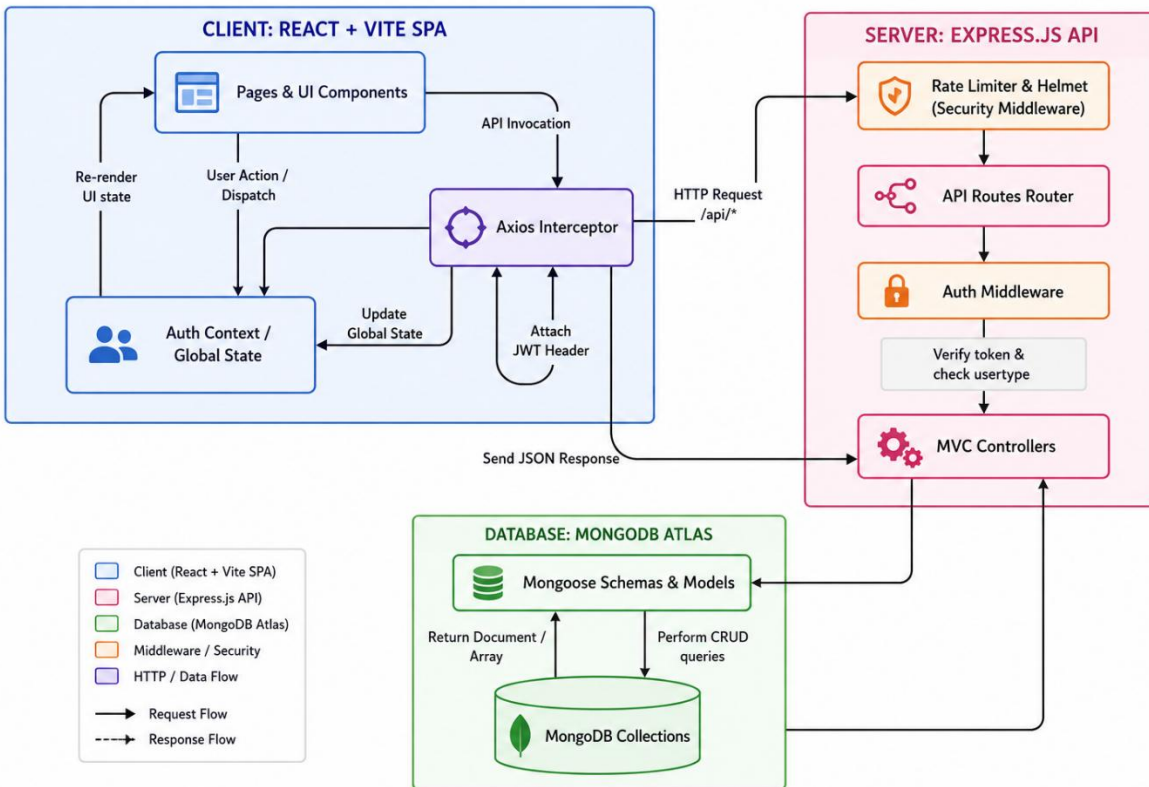


Figure 1: ShopEZ architecture and request/response data flow across the React client, Express API, and MongoDB Atlas database.

Architecture summary:

ShopEZ follows a classic three-tier MERN architecture. The client tier (React 18 + Vite) renders pages and components, manages global auth state via Context API, and uses an Axios instance that automatically attaches the JWT to every outgoing request. The server tier (Node.js + Express 5) is organized in MVC style: incoming requests first pass through Helmet and rate-limiting middleware, then an authentication middleware verifies the JWT and checks the user's role, before routes hand off to controllers that contain the business logic for users, products, cart, orders, and admin operations. The database tier is a MongoDB Atlas cluster accessed through Mongoose schemas, storing Users, Products, Orders, Cart, and Admin (banner/category) collections. Responses travel back through the same layers to update the React UI in real time.